

Review: *A uniqueness theorem for the container derivative*
§6 solidity converse and the $\mathbb{N} \cdot y$ sharpness claim

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Check script: <1 s Python enumeration, reproduced in Appendix A

Verdict

Error, not a gap. The arithmetic of §6 is fine. The structure it is applied to is not the one named in the theorem. Lemma 11 computes D_2 and P' in the Weil-algebra model $\mathbb{N}[\varepsilon_1, \varepsilon_2]/(\dots)$, with ε -truncation and “ $0 \cdot x = 0$ ” absorption, but asserts that the computation takes place in $(\text{Poly}, \triangleleft, y)$. Computed honestly in Poly, $D_2 = (1+n)^2 \cdot y$ and $P' = (1+n) \cdot y$. Universality then gives $(1+n)^2 = 1+n$, which kills $n = 1$, and with it ∂ . The underlying cause is upstream (Finding 1): $T(p) = p + y p'$ is *not* of the form $(-) \triangleleft D$ for any polynomial D . So Theorem 1 is false as stated. I do not accept §6. The sharpness paragraph also overclaims (Finding 4).

Findings

F1. ∂ is not representable in $(\text{Poly}, \triangleleft, y)$. (p. 4, ll. 178–183; Thm 1 p. 2; Thm 16 p. 9.) If $T = (-) \triangleleft D$ then $T(y) = y \triangleleft D = D$. For the dual-number structure $T(y) = y + y \cdot 1 = 2y$, so this forces $D = 2y$. But

$$y^2 \triangleleft 2y = (2y)^2 = 4y^2 \neq 3y^2 = y^2 + y \cdot (y^2)', \quad y^3 \triangleleft 2y = 8y^3 \neq 4y^3.$$

Substitution has no “mod ε^2 ”. The polynomial $y + \varepsilon v$ is just $2y$, and $p(2y)$ keeps every cross term. The left-hand convention fails as well: $D \triangleleft p = 2p$. The sentence “ $(-) \triangleleft D$ is exactly the dual-number substitution T of [11]” (p. 4, l. 179) is the false step. The dual-number T is base change along $\mathbb{N} \rightarrow \mathbb{N}[\varepsilon]/\varepsilon^2$, which is a tensoring in a category of $\mathbb{N}$ -semimodules or $\mathbb{N}$ -algebras, not a right $\triangleleft$ -tensoring in Poly.

F2. Lemma 11 computes in the wrong category. (p. 7, ll. 350–378.) The words “We work in the based setting: ε -grading modulo $\varepsilon^2 \dots$ Weil-algebra model” (l. 351) silently replace Poly by $\mathbb{N}$ -algebras. The three counts mix the two settings:

- $D \triangleleft D$: $(1+n)^2$. This is correct in both settings (true on the linear layer).
- $D_2 = D \times_y D$: the paper gets $1 + 2n$. In Poly, a pullback of linear polynomials over y has position set $S_1 \times_1 S_2$, and its direction set is the pushout $1 \leftarrow 1 \rightarrow 1 = 1$. So $D_2 = (1+n)^2 \cdot y$.

Over a one-shape base the fibre product *is* the product. The sentence “creates no product of the two infinitesimals” (l.366) holds only in the Weil algebra, where $\varepsilon_1\varepsilon_2$ is killed by the quotient.

- P' : the paper gets $1+n+n^2$. In Poly a morphism of linear polynomials is a function on shapes. $\hat{p}$ sends every shape of D to the single shape of y , so $\hat{p} \triangleleft D : (a, b) \mapsto b$ on positions of $D \triangleleft D$. The preimage of $\text{im } \hat{z} = \{0\}$ is $\{(a, 0)\}$, so $P' = (1+n) \cdot y$. With the other convention, $D \triangleleft \hat{p}$, the count is the same. “Kills the mixed term” (l.370) uses the absorption $0 \cdot \varepsilon_{\text{in}} = 0$, and Poly has no such absorption.

The enumeration in Appendix A gives, for $n = 0, \dots, 4$: Poly $(|D_2|, |P'|) = (1, 1), (4, 2), (9, 3), (16, 4), (25, 5)$; Weil $(1, 1), (3, 3), (5, 7), (7, 13), (9, 21)$. Universality holds in Poly only at $n = 0$, and in the Weil model only at $n \in \{0, 1\}$. This is consistent with F1: $(-) \triangleleft 2y$ fails the universality axiom, so it is not a tangent structure at all.

F3. The rest of §6 is sound *conditionally*. (p.8, ll.381–398.) This answers your question (a) directly.

- The D is fixed, so n is fixed. Evaluating the natural iso $T_2 \cong V$ at the single object y gives one equation for that n . That is all you need and all you use, so there is no “for all n ” issue.
- Cancellation in $\mathbb{N}$ is correct: $1 + 2n = 1 + n + n^2 \Rightarrow n = n^2 \Rightarrow n \in \{0, 1\}$.
- Lemma 10’s reduction $T_2 = (-) \triangleleft D_2$, $V = (-) \triangleleft P'$ is plausible for D linear, and I did not find a problem with it.

So the converse is correct for the Weil-model representable structures. But those are exactly Lanfranchi–Lemay’s setting with $R = \mathbb{N}$ and $D = W$ an $\mathbb{N}$ -algebra, not Poly. The “one new idea” (“ranks add, never subtract”) then has nothing Poly-specific left in it. The additive count $1 + n + n^2$ is just the $\mathbb{N}$ -basis count of $\ker(W \otimes W \rightarrow W)$ -over-zero, and it works over any rig with a free W .

F4. Sharpness: Prop. 9 shows the method fails, not that the theorem fails. (p.6, ll.293–310; abstract l.26; Rem. 15 p.8; §9 l.502.) This answers question (b).

- (i) “Consequently unqualified uniqueness over all of Poly is false” (l.297) is a non sequitur. A solid part is not a tangent structure. Your forward direction (§7) is only established for $M \in \{0, y\}$, and you say it “never has to manufacture a third structure”. To claim falsity you must exhibit $(-) \triangleleft \mathbb{N}y$ with all its tangent axioms. As it stands, the correct statement is: *the rank argument does not exclude $\mathbb{N} \cdot y$* .
- (ii) Solidity is about the structure map μ , not about the existence of a bijection $\mathbb{N} \times \mathbb{N} \cong \mathbb{N}$. Cantor pairing shows only that the ranks admit an iso. The $40 \times 40 / 12 \times 12$ “injectivity windows” add nothing, because $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$ is standard.
- (iii) $\mathbb{N} \cdot y \notin \text{SPoly}$, and $(-) \triangleleft \mathbb{N}y$ does not even preserve SPoly. For example, $y \triangleleft \mathbb{N}y = \mathbb{N}y$. So the hypothesis is excluded by the ambient category, not by a rank argument. Please say so. Also separate “finite support” (finitely many shapes) from “finitary” (finite position sets). $\mathbb{N} \cdot y$ is finitary but not finite-support.
- (iv) An interesting by-product of F2: in honest Poly the universality ranks are $|D_2| = \kappa^2 = \kappa = |P'|$ for every infinite κ . So infinite rank passes the test that kills $n = 1$. Remark 15’s “ $1 + 2\kappa = \kappa = 1 + \kappa + \kappa^2$ ” is correct cardinal arithmetic but in the wrong model.

F5. Linearity of M is asserted, not proved. (p.4, ll.198–203.) Poly is not pointed: 0 is initial and 1 is terminal. So “ $M = \ker p$ ” and the splitting $D = y \oplus M$ are not available as written.

The argument “ εy^2 has no vector-space fibre” is heuristic. In fact the additive-bundle axiom alone does not bound n : any finite commutative monoid on $1 + n$ with unit the base shape gives $D_2 \rightarrow D$. Please either prove linearity from the axioms or make it a hypothesis. The Lean boundary statement (p.9) already lists it as a “cited framework fact”. Cited from where?

F6. Minor.

- In Rem. 14 the claim “solidity coincides with indecomposability” is unsupported. T_r for $D = y + \varepsilon v$ has D_r of Weil rank $1 + r$, which is not the object $y + \sum \varepsilon_i v_i$ viewed as a standalone D .
- The Lean certificate (pp.6, 9) certifies $1 + 2n = 1 + n + n^2 \iff n \leq 1$. That is correct and irrelevant to F1 and F2, because the error is in which numbers go into the equation. Please do not cite it as evidence for §6.

Suggested fixes

1. **Change the ambient category.** Restate the theorem for representable tangent structures $T = (-) \otimes W$ on $\mathbb{N}$ -semimodule / $\mathbb{N}$ -algebra objects, i.e. on *coefficients* of polynomials (polynomials with coefficients in a rig R , with T given by base change $R \rightarrow R \otimes W$). That is where your D_2 , P' counts are true, and §6 then goes through as written. Expect a referee to ask what is new over Lanfranchi–Lemay with $R = \mathbb{N}$. The honest answer is “free modules over a rig, no PID”, which is real but small. Drop “(Poly, $\triangleleft$, y)” throughout.
2. **Or keep Poly and report the true answer.** If you insist on right $\triangleleft$ -tensoring in Poly, the finite-rank classification by your own method is $\{\text{id}\}$ *only*: $(1 + n)^2 = 1 + n \iff n = 0$. ∂ is then a non-representable tangent structure. That is a publishable-sounding sentence in its own right, but it is the opposite of Theorem 1.
3. **Rephrase sharpness** as “the rank arguments of §§4 and 6 do not exclude $M = \mathbb{N} \cdot y$; whether $(-) \triangleleft \mathbb{N}y$ (or its Weil analogue) underlies a tangent structure is open.” Do this unless you construct one. Delete “unqualified uniqueness over Poly is false” from the abstract, l.297 and §9.
4. Prove or hypothesise linearity of M (F5).
5. Fix the companion note [11] at the same point (“ $(-) \triangleleft D$ is exactly the dual-number substitution”) if it makes the same identification.

A Check script (summary)

$T(p) = p + yp'$ vs. $p \triangleleft 2y$: they agree on 1 and y ; on y^2 , y^3 , $y^2 + 1$ they give $3y^2$ vs. $4y^2$, $4y^3$ vs. $8y^3$, $3y^2 + 1$ vs. $4y^2 + 1$. Linear pullbacks are computed as fibre products of shape sets. $P' = \{(a, b) \in D \triangleleft D : b = 0\}$, with the same count for $a = 0$.

n	Poly $ D_2 $	Poly $ P' $	iso?	Weil T_2	Weil V	iso?
0	1	1	yes	1	1	yes
1	4	2	no	3	3	yes
2	9	3	no	5	7	no
3	16	4	no	7	13	no
4	25	5	no	9	21	no